Feasibility Study: Rooftop Solar PV for a Business Centre in Bishkek (Huawei, 3 Options)

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Executive summary

For a typical 8-storey business centre in Bishkek, the best option is a 61 kWp grid-tied rooftop PV system on a Huawei SUN2000-50KTL-M3 inverter with zero export. CAPEX is $40k (KGS 3.5M), IRR 13.4%, simple payback 7.9 years, and the plant covers 34% of the building's consumption. At a 14% discount rate NPV is close to zero (KGS −0.14M). A battery, as installed at the Orion-2 business centre, pays off only if uninterrupted power earns about KGS 940k per year on top of energy savings.

	Option 1. Grid-tied 61 kWp	Option 2. Hybrid 61 kWp + 215 kWh	Option 3. Grid-tied 122 kWp with export
CAPEX	$40.1k	$124.0k	$79.7k
NPV (14%, 25 years)	KGS −0.14M	KGS −8.56M	KGS −1.22M
IRR	13.4%	−0.7%	11.6%
Simple payback	7.9 years	none within 25 years	8.9 years
Share of building load	34%	36%	51%
Licence	not required	not required	required (> 15 kW)

The study is based on the current commercial tariff of KGS 4.33/kWh and on building assumptions (250 MWh per year). Assumptions are flagged in each section and incentives are in section 7, open questions in section 8. This is a learning project and will be refined.

1. Site and benchmark

The reference site is a typical 8-storey business centre in Bishkek with a flat roof, banks and office tenants. The benchmark is a real installation: the Orion-2 business centre (178 Panfilov St).

What is known about the benchmark (facts, January 2026):

Parameter	Value	Source
Site	Orion-2 business centre, 8 storeys, 178 Panfilov St; developer: Orion construction company	news report of 8 January 2026, provided by the author
PV plant	50 kW, on the roof	same
Battery	“200 kW” in the courtyard — most likely a unit error in the article: capacity is measured in kWh	same
Claimed effect	grid consumption “reduced to 30%”, up to 50% savings in summer	same
Similar pilot	commercial site in Bishkek: ~50 kW PV + ~200 kWh BESS + EMS, Korean equipment, supported by the Ministry of Economy	24.kg, January 2026

Not verified: whether Orion-2 and the 24.kg pilot are the same site. The parameters match, but the 24.kg article gives no address. “Reduced to 30%” is ambiguous: it may mean “by 30%” or “to 30% of the previous level”. The model assumes “by 30%”.

Choice of building, roof and location (rationale):

- **Building type: business centre.** Peak demand (lighting, air conditioning, IT) coincides with peak solar output on weekdays, so most of the yield is consumed on site. Bank tenants need uninterrupted power, which is an argument for a battery.
- **Roof: flat, accessible.** Modules sit on ballasted frames tilted 15–20° to the south, with no roof penetrations. The optimal tilt for Bishkek is 37° (profileSOLAR), but a lower tilt on a flat roof means less inter-row shading and lower wind load. At 8 storeys there is no shading from neighbouring buildings (to be checked on site).
- **Battery: in the courtyard, on its own foundation.** A LUNA2000-215 cabinet weighs about 2.8 t, so it cannot go on the roof without strengthening the slab. Orion-2 made the same choice.

Building assumptions (to be replaced with real data): annual consumption 250,000 kWh; operating hours on weekdays 08:00–19:00, Saturday a short day; usable roof area for modules 600–1,000 m². Consumption was chosen so that a 50 kW plant gives about the 30% effect claimed for Orion-2.

2. Regulatory framework of the Kyrgyz Republic

Key finding: a PV plant above 15 kW that sells surplus to the grid needs a licence. Options 1 and 2 are therefore designed for zero export, while option 3 includes a licence and a surplus sales contract.

Document	What matters for the project	Status
Law of the Kyrgyz Republic "On Renewable Energy Sources" No. 283 of 31.12.2008	Core law on renewable energy (RES)	in force, as amended
Amendments to the RES law, April 2026	Solar and wind RES operators must install storage of at least 30% of installed capacity. The microgeneration tariff is linked to end-consumer prices. The buy-back coefficient was cut from 1.3 to 1.0. Surplus is bought by the single buyer at tariffs set by the Cabinet of Ministers	signed; law number not found
Presidential Decree No. 212 of 16.06.2026	Temporary special regime for RES from 1 July 2026 to 30 June 2027: government bodies and municipalities must review RES documents as a priority (Economist.kg)	in force since 29.06.2026

Document	What matters for the project	Status
Regulation on rooftop PV (Ministry of Construction + Ministry of Energy)	Up to 15 kW: no licence needed to sell surplus; above 15 kW a licence is required. Required: roof load-bearing assessment, grid connection specifications from the National Electric Grid of Kyrgyzstan (NESK), design documentation. Structures rated for wind up to 45 m/s and snow load. Fire passages up to 1.2 m, grounding, remote shutdown. Bidirectional meter and a contract with the utility, which must sign it within 14 working days. The only permit needed is an urban planning clearance. End-of-life disposal is mandatory	draft as of 30.04.2026; adoption not verified
Medium-Term Tariff Policy 2025–2030 (Cabinet Resolution No. 227 of 24.04.2025, amended by No. 589 of 28.08.2026)	Commercial tariff from 1 May 2026: KGS 4.33/kWh excluding taxes (Economist.kg). Under the Ministry of Energy schedule the tariff rises every May: +33 tyiyn in 2027, +39 in 2028, +47 in 2029, +57 in 2030, after which it covers cost (24.kg). These increments are stated for the residential tariff; no schedule for commercial consumers was found	in force; amendments from 14.09.2026

Implementation steps for the developer (compiled from the draft regulation and not verified in this form):

1. Technical survey of the roof and load-bearing calculation.
2. Grid connection specifications from NESK or the distribution company.
3. Design documentation (electrical, structural, fire safety) and urban planning clearance.
4. Installation, commissioning, acceptance by the utility.

5. For option 3 additionally: licence, bidirectional meter, surplus sales contract. This mechanism has already been tested in an EBRD pilot: five microgeneration sites export power to the grid (Economist.kg, August 2026).

Open legal questions:

- What is the surplus buy-back price for legal entities? Households are paid the average residential tariff of the previous year, KGS 1.45/kWh in 2026. This figure comes from the search snippet of a Tazabek article; the article is paywalled and could not be confirmed. The model uses KGS 1.45 as a conservative assumption.
- Does the “30% storage” requirement apply to rooftop plants for self-consumption? The publication describes it for “RES operators”. The Minister of Energy referred to it for utility-scale plants (Globuc). Needs checking against the text of the law.
- Has the rooftop PV regulation been finally adopted, and in which wording?
- Taxes: 10% profit tax and the depreciation period in the model are assumptions, not checked against the Tax Code of the Kyrgyz Republic.

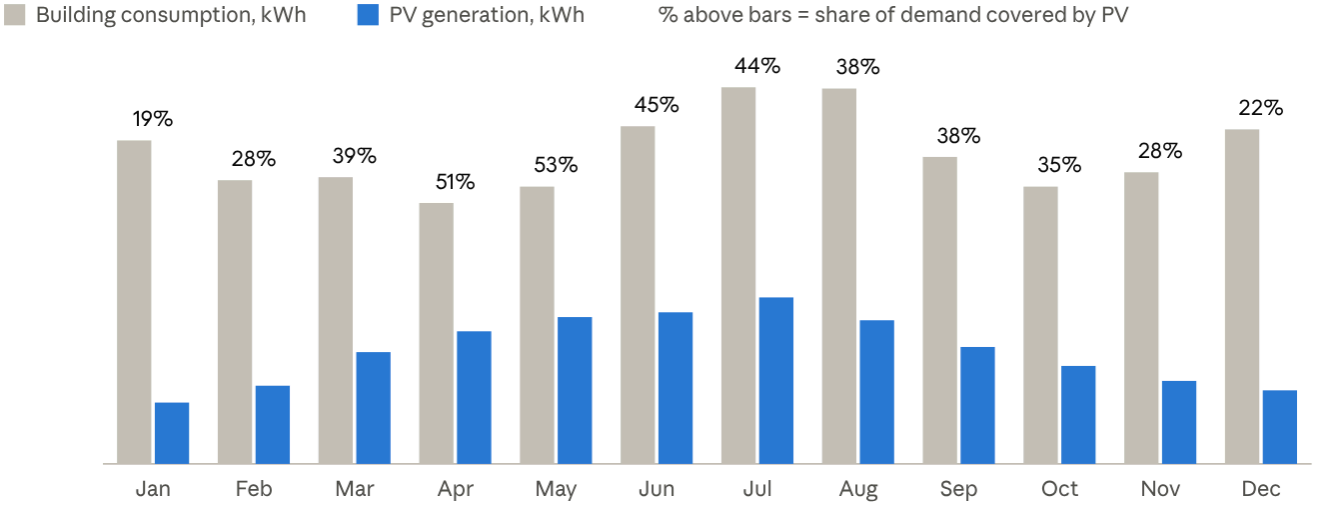
3. Solar resource and yield

The study assumes a specific yield of 1,500 kWh per kWp per year. Benchmarks: PVGIS gives 1,681 kWh/kWp at a 35° tilt and Global Solar Atlas 1,484 kWh/kWp (figures from the search snippet of a Tazabek article of 29.09.2026; the article is paywalled and could not be confirmed when opened). Independent cross-check: the NASA POWER seasonal data below give about 1,770 kWh/kW before system losses (author’s calculation). On a flat roof tilted 15–20° the yield is below optimum, so a value near the lower estimate was chosen.

Average daily yield per 1 kW by season: summer 6.95, spring 5.72, autumn 4.09, winter 2.56 kWh (profileSOLAR, NASA POWER data). In winter the plant produces almost 3 times less than in summer. This coincides with the winter power deficit in the Kyrgyz grid: PV does not replace the grid in winter.

PV covers about 20% of demand in winter and more than half in spring

Option 1 (61 kWp): 91,350 kWh per year, 37% of annual consumption (34% is used on site; weekend surplus is curtailed)



Author's model (model.py): generation profile from NASA POWER / profileSOLAR seasonal data, scaled to 1,500 kWh/kWp; consumption is an assumption of 250,000 kWh/year

Module degradation in the model is 0.4% per year (assumption for N-type TOPCon). The hourly profile is built from an averaged day for each month, without cloud cover, so the self-consumption share may be slightly overstated. Next step: replace the profile with an hourly PVGIS series for the site coordinates.

4. Huawei component market

No official Huawei FusionSolar distributor in Kyrgyzstan could be found in open sources, and there are no public local prices. Prices in the model are therefore based on EU benchmarks and cross-checked against Bishkek retail. A real project needs requests for quotation (RFQ) from regional distributors.

Components for a commercial (C&I) site and price benchmarks:

Component	Key parameters	Price benchmark	In the model
Inverter SUN2000-50KTL-M3	50 kW AC, 400 V 3-phase, 72.2 A (max 79.8 A); DC up to 1,100 V, MPPT 200–1,000 V; 4 MPPT / 8 inputs, 30 A per MPPT, 20 A per input, 40 A short-circuit per MPPT; max efficiency 98.5%; arc fault protection; operating range −25 to +60 °C; 49 kg	€3,564 incl. VAT in the Netherlands (Combishop) ≈ $3,300–3,450 excl. VAT	$3,400 + 10% logistics
Battery LUNA2000-215-2S10	215 kWh, 108 kW AC, battery voltage 648–864 V DC, ~2.8 t (Search4Solar); 0–100% depth of discharge, IP55 (Libra Energy); 0.5C, 91.3% round-trip efficiency, −30…+55 °C — per the datasheet of the sister model LUNA2000-215-2S12 (Chinese version); recommended PV array 115 kWp (Davis & Shirtliff)	EU: €50.3k and €69.9k, VAT basis not stated (Search4Solar); €75.9k incl. VAT (Combishop); €79.8k incl. VAT, lead time 60–90 days (7sun)	$64,000 + 10%
SmartLogger3000 + DTSU666-HW meter	control and monitoring, export limitation, peak shaving without an external EMS; up to 20 LUNA cabinets per logger (Libra Energy)	not found	$1,000 (assumption)

Component	Key parameters	Price benchmark	In the model
Modules (not Huawei)	TOPCon 580 W, typical datasheet	Bishkek retail about $0.30/W (lalafo listing, October 2026: 500 W for KGS 13,000; from a search snippet, the page did not open when checked)	$0.22/W wholesale + 10% (assumption)
Turnkey PV (for cross-check)	grid-tied kits 2–10 kW	KGS 58–66k per kW ≈ $660–760 (220.kg prices from the search snippet of a Tazabek article; the article is paywalled)	model result: $654–659/kWp

Currency conversion: KGS 87.45 per USD — the rate shown in the header of 24.kg (the page gives no date for the rate). Euros were converted approximately at €1 ≈ $1.13–1.17: different 24.kg pages show the euro at KGS 98.34 to 102.07. For EU prices including VAT, tax was removed at the seller's country rate (Netherlands 21%, author's assumption). All prices in the model exclude VAT.

Huawei's position in the region (facts): the region is covered by Huawei Digital Power for the Middle East and Central Asia. In May 2024, together with Photomate, it presented products for the Central Asian distribution market in Tashkent (Spot.uz, sponsored content). Photomate describes itself as a Huawei Value Added Partner (Photomate). According to the ENF directory, Soventus, a distributor active in Kyrgyzstan, carries a competitor — SOFAR inverters and Jinko modules (from a search snippet; the ENF page did not open when checked).

Competition and weak points (author's assessment; market shares not verified):

- Main competitors in C&I: Sungrow, SOFAR, Deye, Growatt, Solis; SMA in the premium segment.
- Huawei strengths for a developer: one ecosystem from one brand (inverter + battery + SmartLogger + FusionSolar cloud), arc fault protection (important for roof fire safety), a brand banks recognise.
- Weak points: higher price than Deye/SOFAR, an opaque sales channel in Kyrgyzstan (no public distributor or service centre), long battery lead times.
- Who to sell to: developers of business and shopping centres, banks, commercial sites with the KGS 4.33 tariff and a daytime load profile. Separately, government buildings: the budget-sector tariff is even higher at KGS 4.62. The Green Energy Fund is already

working on rooftop PV for administrative buildings in Bishkek (Tazabek, November 2026, provided by the author).

5. Three options: equipment and electrical design

All options are built from one standard block: 7 strings of 15 modules per SUN2000-50KTL-M3 inverter. Option 2 reproduces the Orion-2 configuration on Huawei equipment.

	Option 1. Grid-tied	Option 2. Hybrid (like Orion-2)	Option 3. Extended
DC / AC capacity	60.9 kWp / 50 kW	60.9 kWp / 50 kW	121.8 kWp / 100 kW
TOPCon 580 W modules	105 (7 strings × 15)	105	210 (14 strings × 15)
Inverters	1 × SUN2000-50KTL-M3	1 × SUN2000-50KTL-M3	2 × SUN2000-50KTL-M3
Battery	—	LUNA2000-215-2S10, 215 kWh, in the courtyard	—
Control	SmartLogger3000 + meter, zero export	same + automatic transfer switch (ATS) for island mode	SmartLogger3000 + bidirectional meter
Surplus handling	curtailed (no licence)	charges the battery	sold to the grid (licence)
Roof area (estimate)	~450–550 m²	~450–550 m² + 2×3 m pad in the courtyard	~900–1,100 m²
Generation, year 1	91,350 kWh	91,350 kWh	182,700 kWh
Used on site	84,589 kWh (93%)	90,762 kWh (99%)	127,765 kWh (70%)
Surplus	6,761 kWh curtailed	~0 (battery cycle losses)	54,935 kWh exported
Share of building load	34%	36%	51%

Roof area is the author's estimate: a module is ~2.6 m², with a factor of ~1.7–2 for row spacing and fire passages. Option 3 is only feasible if the roof offers about 1,000 m² of free area.

String voltage check (15 modules). Module parameters are typical for TOPCon 580 W and must be confirmed against the datasheet of the chosen brand: Voc 51.6 V, Vmp 43.2 V, Isc 14.3 A, Imp 13.4 A, temperature coefficient of Voc −0.25%/°C, Vmp −0.30%/°C. Inverter limits are from the Huawei datasheet.

Check	Calculation	Result	SUN2000-50KTL-M3 limit
Voc at −30 °C (assumed minimum for Bishkek)	15 × 51.6 × (1 + 0.0025 × 55)	880 V	⩽ 1,100 V — OK
Vmp at cell temperature +70 °C	15 × 43.2 × (1 − 0.003 × 45)	561 V	within MPPT 200–1,000 V — OK; full-load range not confirmed*
Vmp at −10 °C	15 × 43.2 × (1 + 0.003 × 35)	716 V	within MPPT — OK
Current per MPPT (2 strings)	2 × 13.4 / 2 × 14.3	26.8 A / 28.6 A	30 A operating / 40 A short-circuit — OK
Current per input (1 string)	13.4 / 14.3	13.4 A / 14.3 A	20 A — OK
Strings per MPPT	7 strings on 4 MPPT	2 + 2 + 2 + 1	8 inputs — OK
DC/AC ratio	60.9 / 50	1.22	typically 1.1–1.3 (author's estimate)

* The English Huawei datasheet that was opened does not state the full-load MPPT range. According to search snippets, the German datasheet gives 530–800 V at 400 V AC and a Ukrainian retailer lists 380–850 V. In both cases 561–716 V is within range. The inverter operates down to −25 °C, while Bishkek winters can be colder (author's estimate), so the inverter is better installed in a top-floor plant room or under a canopy.

AC side and connection. Inverter output is 400 V, 3 phases + N + PE, rated current 72.2 A, maximum 79.8 A. Proposed breaker: 100 A per inverter (assumption; cable cross-section depends on the route length from the roof to the main switchboard and is sized during design). The plant connects to the building's 0.4 kV main switchboard downstream of the incoming meter. The DTSU666-HW meter at the incomer lets the SmartLogger limit export (options 1 and 2) or track power flow (option 3). Option 3 also needs a bidirectional revenue meter from the utility. In option 2 the battery connects on the AC side at 400 V

through its built-in converter (battery voltage 648–864 V DC). Operation during a grid outage needs an ATS and a dedicated busbar for critical loads (banks, server rooms, emergency lighting, lift). The model reserves 30% of capacity (~65 kWh) for backup power.

6. Financial model

At a 14% discount rate no option has a positive NPV. Option 1 is very close to break-even: IRR 13.4%, simple payback 7.9 years. The battery (option 2) does not pay off on energy savings: its value lies in uninterrupted power.

Base assumptions

Parameter	Value	Basis
Commercial tariff	KGS 4.33/kWh excl. taxes	fact, from 01.05.2026
Tariff growth	8%/yr to 2030, then 5%	assumption
Surplus buy-back price (option 3)	KGS 1.45/kWh	assumption (household rate)
Discount rate	14%, nominal in KGS	National Bank policy rate 12% (Economist.kg, 27.10.2026) + 2 pp
Exchange rate	KGS 87.45 per USD, constant	assumption; risk tested in sensitivity
Profit tax / depreciation	10% / 10 years straight-line	assumption, to be checked against the Tax Code
O&M	1% of PV CAPEX + 0.5% of battery price per year, +5%/yr	assumption
Replacements	inverters in year 12; batteries in year 13 at 60% of today's price	assumption
Horizon	25 years, commissioning in 2027	assumption

CAPEX by option, $ excl. VAT

Item	Option 1	Option 2	Option 3
Modules (incl. logistics)	14,738	14,738	29,476

Item	Option 1	Option 2	Option 3
SUN2000-50KTL-M3 inverters	3,740	3,740	7,480
SmartLogger + meter	1,100	1,100	1,100
Mounting, cabling, protection	9,135	9,135	18,270
Design, installation, commissioning	7,308	7,308	14,616
Roof survey	1,500	1,500	1,500
LUNA2000-215 battery	—	70,400	—
ATS + battery installation	—	8,000	—
Licence, meter, contract	—	—	2,000
Contingency 7%	2,626	8,114	5,211
Total	**40,147**	**124,035**	**79,653**
Per kWp	659	2,037	654

Metrics (25 years, 14% discount rate)

Metric	Option 1	Option 2	Option 3
CAPEX, KGS M	3.51	10.85	6.97
Net cash flow, year 1, KGS k	360	357	614
NPV, KGS M ($k)	−0.14 (−1.7)	−8.56 (−97.9)	−1.22 (−13.9)
IRR	13.4%	−0.7%	11.6%
Simple payback	7.9 years	none within 25 years	8.9 years
Discounted payback	over 25 years	none	over 25 years
ROI over 25 years (nominal)	349%	−11%	285%
Profitability index (PI)	0.96	0.21	0.83
LCOE, KGS/kWh generated	6.46	20.74	6.41
Levelised grid tariff over 25 years, KGS/kWh	7.02	7.02	7.02

How to read this: the LCOE of options 1 and 3 is below the levelised grid tariff, so own generation is cheaper than buying from the grid. But the margin is thin. In option 1 it is eaten by curtailed weekend surplus and tax. In option 3, 30% of output is sold at KGS 1.45 instead of 4.33. ROI looks high because it ignores the time value of money; decisions are better based on NPV and IRR.

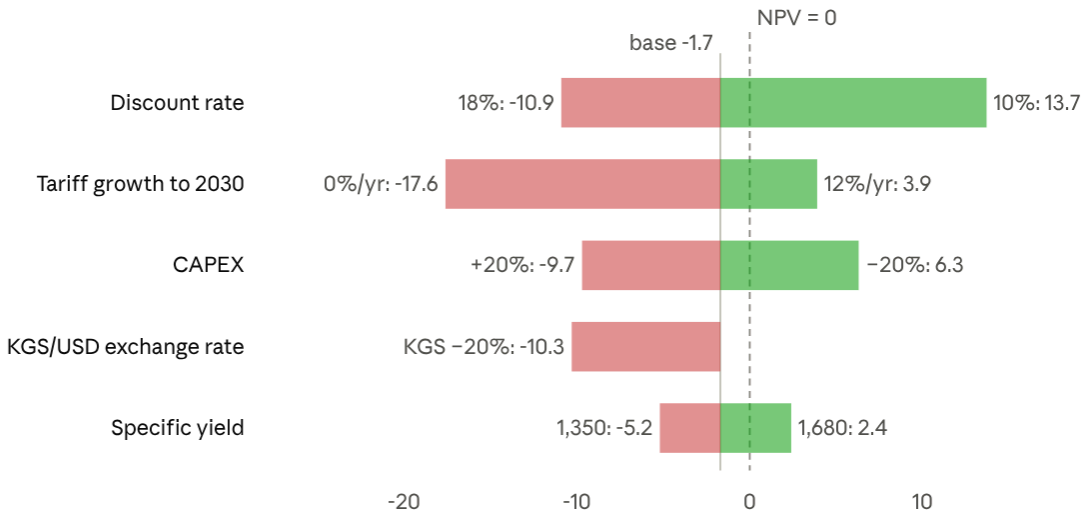
Model limitations found during review (the arithmetic was recomputed independently and matched to the som):

- **Smoothed generation profile.** The averaged day peaks at about 0.68 kW per kWp, while the real peak at a clear summer noon is higher (author's estimate 0.8–0.85). The model therefore misses inverter clipping (DC/AC 1.22) and part of the midday surplus, so self-consumption and savings are probably slightly overstated. Fix: hourly PVGIS series.
- **The battery in option 2 is under-used.** It delivers about 6,200 kWh per year to the building, roughly 29 full cycles. Huawei recommends about 115 kWp of PV for a LUNA2000-215; here it is 61 kWp, and almost all output goes straight to the load. Grid charging and peak shaving are not modelled: no time-of-use tariff or demand charge for commercial consumers was found.
- **8% tariff growth is an assumption.** The official increment schedule is stated for households, where growth is about 20% per year. The commercial tariff of KGS 4.33 is already above average cost (Ministry of Energy forecast for 2026: KGS 2.99/kWh, Economist.kg), so assuming the same growth for it would be imprudent.

What uninterrupted power must be worth (option 2). For option 2 to reach zero NPV, the battery must earn about KGS 940k ($10.8k) per year on top of savings, at 2027 prices growing 5% per year. This could be a rent premium for tenants with guaranteed power, or avoided diesel generator and fuel costs. The question for the developer: are tenants willing to pay about KGS 78k per month for outage-free power?

Option 1 turns positive if capital costs less than 14% or tariffs grow faster than 8%/yr

25-year NPV, $ thousands; one factor changed at a time, others at base. Green = better than base, red = worse



Author's model (model.py), sensitivity analysis of option 1; KGS 87.45 per USD

Formulas used by the model (CF — net cash flow of year t after O&M, replacements and tax; r = 14%; E — generation):

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NPV = -CAPEX + \sum_{t=1}^{25} \frac{CF_t}{(1+r)^t}, \qquad \sum_{t=0}^{25} \frac{CF_t}{(1+IRR)^t} = 0, \qquad LCOE = \frac{CAPEX + \sum_t \frac{O\&M_t + Repl_t}{(1+r)^t}}{\sum_t \frac{E_t}{(1+r)^t}}
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7. Incentives

The incentive with the biggest impact is a concessional "green" loan. If capital costs 10% instead of 14%, NPV becomes +KGS 1.20M for option 1 and +KGS 1.06M for option 3. The import VAT exemption cuts CAPEX by 6–9% if the developer cannot recover VAT. A tax holiday changes almost nothing.

Incentive	Terms	Relevance to the project	Source
VAT-free import of RES equipment	List approved by the Cabinet of Ministers (resolution of 4 October 2021). Requires confirmation of intended use by the authorised energy body. Resolution No. 272 of 22.04.2026 expanded the list: electrotechnical equipment, transformers, cables, control systems. Resolution No. 603 of 08.09.2026 removed one item (pipes for small hydro)	Model prices already exclude VAT. For a developer who cannot recover VAT, the saving is 12% of imported equipment: option 1 — $2.3k (5.9% of CAPEX), option 2 — $11.2k (9.0%), option 3 — $4.6k (5.7%). Whether specific Huawei models are on the list must be checked by HS code	Economist.kg, 2021 · Economist.kg, September 2026
Customs duty	Base EAEU rate is 15% of customs value, but duty on photovoltaic batteries within the EAEU is zero	Modules: probably 0%. Inverters and batteries: rate to be checked by HS code. The model covers all import costs with a single 10% “logistics and customs” allowance	Kaktus.media, April 2025
“Improving Air Quality” concessional loan	For businesses: up to KGS 5M at 10% per year via Eldik Bank, ABank and Bakai Bank. Can be used for solar panels. Term and collateral are not stated in the article	Option 1 (KGS 3.51M) fits fully within the limit. Option 3 (KGS 6.97M) only partly. Option 2 (KGS 10.85M) less than half	Economist.kg, June 2026

Incentive	Terms	Relevance to the project	Source
Green loans of the State Development Bank	$60M from a World Bank concessional loan for green projects of micro, small and medium enterprises, at below-market rates (rate not stated)	An alternative for amounts above KGS 5M if the developer qualifies as an SME	Economist.kg, April 2025
Tax exemption for new RES energy producers	Up to 5 years from commissioning	Most likely not applicable to a self-consumption plant (author's assessment). Even if applied, the effect is small: depreciation already almost eliminates tax in the early years	Kaktus.media, April 2025
RES special regime (Decree No. 212)	From 1 July 2026 to 30 June 2027 government bodies review RES documents as a priority	Speeds up approvals if documents are filed before June 2027	Economist.kg

Incentive scenarios in the model (25-year NPV, KGS M)

Scenario	Option 1	Option 2	Option 3
Base: 14% discount rate, no incentives	−0.14	−8.56	−1.22
Financing at 10%	+1.20	−7.88	+1.06
5-year tax holiday	−0.12	−8.56	−1.21
Financing at 10% + tax holiday	+1.22	−7.88	+1.07

Method: "financing at 10%" is a simplification — the whole project is discounted at the concessional loan rate. This is fair for option 1, which fits entirely within the KGS 5M limit. For option 3 the scenario is optimistic: about KGS 2M would have to be funded at another

rate. Loan term and repayment schedule were not modelled because the source does not give them. IRR does not depend on the discount rate and stays at 13.4% and 11.6%: the project becomes profitable because its return now exceeds the cost of capital.

Why a tax holiday barely helps: for the first 10 years depreciation (KGS 351k per year for option 1) almost equals the profit from savings. Tax in year 1 is about KGS 1k, so there is practically nothing to exempt.

Takeaway for the developer:

1. Apply for an “Improving Air Quality” loan to finance option 1.
2. Before customs clearance, obtain confirmation of intended use for the VAT exemption.
3. File documents for approval before 30 June 2027, while the special regime is in force.

8. Conclusions, risks and open questions

Recommendation: option 1 — a 61 kWp grid-tied PV plant with zero export. It has the best IRR (13.4%) and payback (7.9 years), needs no licence, and matches the scale of Orion-2. The project turns NPV-positive under any of these conditions: financing cheaper than 14%, CAPEX 5–10% below the model (through RFQs), or tariff growth faster than 8% per year.

- **Option 2 (like Orion-2)** is justified only if uninterrupted power can be monetised at about KGS 940k per year. On tariff savings alone the battery does not pay off: there is no time-of-use tariff for commercial consumers that the battery could earn on. The option is worth revisiting if a business time-of-use tariff appears or if the storage requirement is extended to rooftop plants.
- **Option 3** brings more money in absolute terms (KGS 614k in year 1) but less per som invested. At a surplus buy-back price of KGS 3.5 its IRR matches option 1 (13.8%); at KGS 4.33 its NPV turns positive (+KGS 0.34M). The sensible path is to build option 1 with room for expansion and extend once the buy-back price for legal entities is known.

Risks

Risk	Impact	Mitigation
Tariff grows slower than planned	high: with zero growth, IRR of option 1 falls to 5.6%	evaluate the project at zero growth too; do not promise the client a payback based on the tariff plan

Risk	Impact	Mitigation
Som depreciation	medium: equipment is in USD, savings in KGS	fix the contract price, buy in one batch
Roof load-bearing capacity	blocking	survey before procurement; ballast ~15–25 kg/m² (estimate)
Regulatory change (licence, 30% storage, buy-back)	medium	lock in grid connection specifications and the contract before construction; Decree No. 212 special regime runs until 30.06.2027
Huawei service and warranty in Kyrgyzstan	medium	buy through the official channel, put an SLA in the contract
Roof and battery fire safety	high consequences	arc fault protection in the inverter, remote shutdown, fire passages, battery outside the building
Self-consumption overstated in the model	low–medium	use the building's real hourly load profile from its meter

What needs to be clarified to make this a working study

- [ ] Actual building consumption: 12 months of bills and an hourly profile from the meter
- [ ] Roof area and load-bearing capacity, shading from rooftop structures
- [ ] Surplus buy-back price for legal entities and the licensing procedure above 15 kW
- [ ] Whether the "30% storage" requirement applies to rooftop plants for self-consumption
- [ ] Quotations for Huawei equipment from a distributor in Kyrgyzstan or the region (inverter, LUNA2000, SmartLogger)
- [ ] Profit tax rate, depreciation period and RES incentives under the Tax Code of the Kyrgyz Republic
- [ ] The developer's cost of capital (own funds, loan, green credit lines)

Sources

Data collected and re-checked on 6 November 2026. “Search snippet” means the page did not open when checked or is paywalled, and the figure was taken from the search result excerpt. Most sources are in Russian.

- Orion-2: news report of 8 January 2026 (text provided by the author; original source not opened)
- 24.kg — 50 kW PV + 200 kWh BESS pilot
- Law of the Kyrgyz Republic “On Renewable Energy Sources” No. 283 (FAOLEX)
- Economist.kg — amendments to the RES law, April 2026
- Economist.kg — RES special regime under Decree No. 212 · Decree text, spinform
- QAZAQ GREEN — draft rooftop PV regulation
- Economist.kg — tariffs from 1 May 2026 · 24.kg — tariff schedule to 2030 and KGS rate
- Cabinet Resolution No. 589 of 28.08.2026 on tariff policy, spinform
- Economist.kg — electricity cost vs average tariff
- Economist.kg — EBRD microgeneration pilot
- Globuc — Ministry of Energy on the 15 kW threshold and 30% storage
- Tazabek — household PV: PVGIS/GSA, prices per kW, KGS 1.45 buy-back (search snippet; article paywalled)
- profileSOLAR — seasonal yield, Bishkek
- Economist.kg — National Bank policy rate 12%, October 2026
- Huawei — SUN2000-50KTL-M3 datasheet
- LUNA2000-215: Huawei — 2S12 datasheet, Chinese version · Search4Solar — 2S10 parameters · Davis & Shirtliff — recommended PV array · Libra Energy — SmartLogger and scaling
- Prices: Combishop — SUN2000-50KTL-M3 · Combishop, Search4Solar, 7sun — LUNA2000-215 · modules — lalafo listing (search snippet)
- Market: Spot.uz — Huawei FusionSolar Day in Tashkent · Photomate · ENF — Soventus (search snippet)
- Model: model.py (Python, author’s calculation; all assumptions in one block)